Note : This Practic book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.

Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
UNIT - 8 - Secondary Storage and input Output Management								
TOPIC NAME:- I/O Devices, Device Controllers, Device Drivers, Disk Hardware Structure, Disk Scheduling(FCFS,SSTF,SCAN,C-SCAN,LOOK,C-LOOK)								
282	8	The _____keeps state information about the use of I/O components.	C	1	CPU	OS	Kernel	Shell
283	8	When hardware is accessed by reading and writing to the specific memory locations, then it is called _____	D	1	port-mapped I/O	controller-mapped I/O	bus-mapped I/O	memory-mapped I/O
284	8	The heads of the magnetic disk are attached to a _____that moves all the heads as a unit.	C	1	spindle	track	disk arm	pin
285	8	The consists of two key components: the initial startup time, and the time taken to traverse the tracks that have to be crossed once the access arm is up to speed.	C	1	access time	rotational delay	seek time	Transfer time
286	8	On a movable head system, the time it takes to position the head at the track is known as ……	A	1	seek time	rotational delay	access time	Transfer time
287	8	The time disk controller takes for the beginning of the sector to reach the head is known as ……..	B	1	seek time	rotational delay	access time	Transfer time
288	8	The set of tracks that are at one arm position make up a _____	B	1	magnetic disks	cylinders	assemblies	electrical disks
289	8	In policy, when the last track has been visited in one direction, the arm is returned to the opposite end of the disk and the scan begins again	D	1	Last in first out	Shortest service time first	SCAN	Circular SCAN
290	8	What is device driver? Explain its function in brief.		3				
291	8	Explain First Come First Serve and Shortest Seek Time First Disk Scheduling algorithms.		4				
292	8	Enlist any Three Disk Arm Scheduling Algorithms and explain them briefly.		5				
293	8	Briefly describe SCAN.		4				
294	8	Explain various Disk Arm Scheduling Algorithms.		5				
295	8	Explain SSTF and LOOK disk scheduling algorithms with examples.		5				
296	8	Write short note: Direct memory access (DMA)		3				
297	8	Disk requests come in to the disk for cylinders 10, 22, 20, 2, 40, 6 and 38. A seek takes 6 msec per cylinder move. How much seek time is for Closest cylinder next algorithm? Initially arm is at cylinder 20.		5				
298	8	Disk requests come in to the disk driver for cylinders 10, 22, 20, 2, 40, 6, and 38, in that order. A seek takes 6 msec per cylinder moved. How much seek time is needed for 1. First-come, first served. 2. Elevator algorithm. (initially moving upward) In all cases, the arm is initially at cylinder 20.		5				
299	8	Consider the following disk request sequence for a disk with 100 tracks 45, 21, 67, 90, 4, 50, 89, 52, 61, 87, 25. Head pointer starting at 50 and moving in left direction. Calculate head movement for the following algorithms. 1. FCFS 2. SSTF		5				
300	8	Suppose that a disk drive has 200 cylinders. Numbered 0 to 199. The drive is currently serving at cylinder 60 and previous request was at cylinder 35. The queue of pending requests in FIFO order is 70, 140, 50, 125, 30, 25, 160 Starting from the current head position what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for each of following disk scheduling algorithms. a) FCFS b) SSTF c) SCAN d) LOOK		5				
301	8	Suppose Disk drive has 300 cylinders. The current position of head is 90. The queue of pending request is 36,79,15,120,199,270,89,170 Calculate head movement for the following algorithms. 1. FCFS 2. SSTF		5				
302	8	Given the following track requests in the disk queue, compute for the Total Head Movement (THM) of the read/write head and seek time needed for a). First come first served b). Shortest seek time first c). SCAN: 95, 180, 34, 119, 11, 123, 62, 64 Consider that the read/write head is positioned at location 50. A seek takes 5ms per cylinder move.		5				
303	8	Suppose that a disk drive has 200 cylinders, numbered 0 to 199. The drive is currently serving at cylinder 53 and previous request was at cylinder 43. The queue of pending requests in FIFO order is: 98, 183, 37, 122, 14, 124, 65, 67 Starting from the current head position what is the total distance(in cylinders) that the disk arm moves to satisfy all the pending requests for each of the following disk scheduling algorithms. a)FCFS b)SSTF c)SCAN d)LOOK e)C-SCAN f)C-LOOK		5				
304	8	Define seek time and rotational latency. Assume that a disk drive has 200 cylinders, numbered 0 to 199. The drive is currently serving a request at cylinder 100. The queue of pending requests is 23, 89, 132, 42, 189.Calculate seek time for FCFS and SSTF disk scheduling algorithm.		5				
305	8	Suppose that a disk drive has 5000 cylinders, numbered 0 to 4999. The drive is currently serving a request at cylinder 143, The queue of pending requests, in FIFO order, 86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130 Starting from current head position what is total distance (in cylinders) that disk arm moves to satisfy all the pending request for FCFS and SSTF disk scheduling algorithm?		5				
306	8	Consider a disk queue with I/O requests of the following cylinders in their arriving order 6,10,12,54,97,73,128,15,44,110,34,45 The disk head is assumed to be at Cylinder 23 and moving in the direction of decreasing number of cylinders. The disk consists of total 150 cylinders. Calculate the disk head movement using SCAN and C - SCAN scheduling algorithm		5				

UNIT - 9 - Security and Protection Mechanism								
TOPIC NAME:-Security Environment, Protection Mechanism, User Authentication Methods								
307	9	Which of the following is a strong password?	A	1	P@ssw0rd	!augustdelhi	19thAugust88	Delhi88
308	9	_____in an operating system refers to the mechanisms implemented by the operating system to ensure the security and integrity of the system.	A	1	System protection	Access control	Encryption	Firewall
309	9	_____is a software program that monitors and controls incoming and outgoing network traffic based on predefined security rules	B	1	Antivirus software	Firewall	System protection	System updates
310	9	_____is used to protect the system from malware, and other malicious software.	A	1	Antivirus software	Firewall	System protection	System updates
311	9	_____keeps the operating system up-to-date with the latest security patches to prevent known vulnerabilities from being exploited	A	1	System updates	Antivirus software	Firewall	System protection
312	9	_____gives the system access only to authorized users and tackles the system's external threats	A	1	Security	System protection	System updates	Antivirus software
313	9	_____deals with who has access to the system resources and tackles the system's internal threats.	A	1	Protection	System updates	Antivirus software	Security
314	9	_____is a form of malware that downloads on a computer by impersonating a trustworthy program and can get unauthorized access to a system's login information.	B	1	Worm	Trojan Horse	Trap door	Virus
315	9	_____is a sort of malware whose main purpose is to keep operating on infected systems while self-replicating and infecting other computers.	A	1	Worm	Trojan Horse	Trap door	Virus
316	9	_____has the ability to produce duplicate copies that occupy all available resources and prevent any other processes from using them.	A	1	Worm	Trojan Horse	Trap door	Virus
317	9	_____aims to shut down a computer system or network so that its intended users are unable to access it and prevent authorized users from accessing a system.	D	1	Worm	Trojan Horse	Trap door	Denial of Service
318	9	Which mechanism is implemented in protection to control access to system resources?	C	1	Encryption	Certification	Authorization	none
319	9	Netbus, Subseven, Y3K Remote Administration Tool are types of_____	B	1	Worm	Trojan Horse	Trap door	Denial of Service
320	9	Robert Morris is an example of _____	A	1	Internet Worm	Trojan Horse	Trap door	Virus
321	9	Back Orifice, Beast, Zeus, The Blackhole Exploit Kit are example of_____	B	1	Worm	Trojan Horse	Trap door	Denial of Service
322	9	Write short note on Design Principles of Security		3				
323	9	Explain Domain Protection mechanism in detail		3				
324	9	Explain Access metrics mechanism.		3				
325	9	What is Access Control list? Explain in brief		3				
326	9	Explain Trojan Horse, Trap Door, Virus and Warms program threats.		3				
327	9	Explain User Authentication Methods		3				
UNIT-10 - Unix/Linux Operating System								
TOPIC NAME:-Role & Function of Kernel, Linux Kernel Structure , Elementary Linux Command and Shell Programming								
328	10	Which of the following is a Unix command to list all files in a directory?	B	1	list	ls	dir	show
329	10	What is the purpose of the chmod command in Unix?	C	1	Change file content	Change file owner	Change file permissions	Change file name
330	10	How do you delete a directory and its contents in Unix/Linux?	D	1	del	remove -f	deleteall	rm -r
331	10	What is the Unix command to display the present working directory?	A	1	pwd	where	home	presentdir
332	10	What is the core of the Unix operating system?	B	1	Shell	Kernel	Command	Script
333	10	What does the grep command do?	A	1	Searches for a string in a file	Deletes a file	Saves a file	Moves a file
334	10	How do you search for a manual page in Unix/Linux?	C	1	info	help	man	guide
335	10	Which command is used to display the content of a file in Unix?	B	1	show	cat	disp	display
336	10	Which of the following is the symbol for piping in Unix/Linux?	D	1	%	*	#	|
337	10	What does the ps command do in Unix/Linux?	B	1	Displays file permissions	Displays process status	Displays current users	Displays system information
338	10	Which of the following is a process control command in Unix/Linux?	A	1	kill	ctrlp	stopproc	ps
339	10	What command is used to change file ownership in Unix/Linux?	B	1	chmod	chown	change	ownfile
340	10	What command is used to copy files in Unix/Linux?	C	1	copy	cop	cp	duplicate
341	10	What does the df command display?	B	1	File size	Disk space usage	Memory usage	CPU usage
342	10	Which of the following file types does Unix/Linux not recognize?	D	1	Regular file	Directory file	Symbolic link file	Executable file
343	10	Explain in brief roles/ functions of shell		3				
344	10	Explain the directory Structure or file system structure of the Unix.		4				
345	10	Write a shell script print multiplication table of given number.		4				
346	10	Write a shell script find out reverse of given number.		4				
347	10	Write a shell script to Check given number is Armstrong or not.		4				
348	10	Write a shell script which will accept a number b and display first n prime numbers as output.		4				
349	10	Write a shell script to check entered string is palindrome or not.		4				
350	10	Write a shell script to read n numbers as command arguments and sort them in ascending order.		4				
351	10	Write a shell script to find the greater number from three number.		4				
352	10	Write a shell script to find the number is perfact square or not.		4				